

Production Networks and Formation of Aggregate Uncertainty

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Abstract

JEL Codes: Keywords:

1 Introduction

(Acemoglu and Azar, 2020)

2 Benchmark Model

In the benchmark model, we assume that all the agents have full information with rational expectations (FIRE). The friction in this economy is the financial friction in the form of borrowing constraint.

Time is discrete and indexed by $t = 0, 1, 2, \dots$. There are representative agents with unit mass who consume a consumption aggregator C_t , make nominal deposits D_t and supplies labour N_t , a representative firm in each sector $i \in \{1, 2, \dots, N\}$ ¹ that produces a differentiated good $y_{i,t}$ using labour $n_{i,t}$ and intermediate input $x_{ij,t}$ from sectors j , competitive financial intermediaries that takes deposits from households and supplies credit to firms, and a monetary authority that sets the nominal interest rate i_t .

Interest rates are dated by when they are set, but need not apply over the same window. The policy rate i_t is set at the beginning of period t and remunerates deposits from the beginning of period $t + 1$ to its end (so deposits D_t earn i_t in period $t + 1$). The intra-period loan rate $i_{i,t}^L$ is set at the beginning of period t and applies from the beginning of period t to its end; it is priced off the predetermined funding rate i_{t-1} that prevails over that same window.

The timeline within a period t is as follows. At the beginning of the period, the monetary authority sets the nominal interest rate i_t . The financial intermediary charges firm i the loan rate $i_{i,t}^L$, and firms borrow, hire labour, and buy intermediate inputs. At the end of the period, productivity realizes, firms produce and sell goods and repay $(1 + i_{i,t}^L) b_{i,t}$, and households consume, collect the maturity payoff $(1 + i_{t-1})$ on deposits D_{t-1} , and choose new deposits D_t .

¹In the following, we will use the sector index i and firm i interchangeably.

2.1 Environment

2.1.1 Households

Households gain utility from a consumption aggregator C_t and disutility from labour supply N_t :

$$U(C_t, N_t) = \log C_t - \frac{\chi}{1 + \varphi} N_t^{1+\varphi}, \quad (2.1)$$

with $\chi > 0$ and $\varphi > 0$, where the consumption aggregator is given by a CES aggregator of the differentiated goods $c_{i,t}$:

$$C_t = \left(\sum_{i=1}^N \rho_i^{\frac{1}{\xi}} c_{i,t}^{\frac{\xi-1}{\xi}} \right)^{\frac{\xi}{\xi-1}} \quad (2.2)$$

with $\rho_i \in (0, 1)$ the demand weight of goods i satisfying $\sum_{i=1}^N \rho_i = 1$ and $\xi > 0$ the elasticity of substitution (EoS) between goods. As a result, the price level is given by

$$P_t = \left(\sum_{i=1}^N \rho_i p_{i,t}^{1-\xi} \right)^{\frac{1}{1-\xi}}. \quad (2.3)$$

where $p_{i,t}$ is the price of good i .

They are subject to a budget constraint:

$$P_t C_t + D_t \leq W_t N_t + \left(1 + i_{t-1}^D\right) D_{t-1} + \Pi_t^y + \Pi_t^f, \quad (2.4)$$

where W_t is the nominal wage, i_{t-1}^D is the nominal interest rate on deposits made in period $t-1$ (paid in period t), Π_t^y is the total nominal profit of all the firms, and Π_t^f is the total nominal profit of the financial intermediary.

2.1.2 Firms

Firm i uses intermediate inputs from other sectors so that the economy has a network, denoted by an $N \times N$ matrix $\Omega_t = (\omega_{ij,t})_{i,j=1}^N$ where the (i, j) -th entry $\omega_{ij,t}$ denotes the expenditure share of firm i on input j . The corresponding quantity of j used to produce i is $x_{ij,t}$. Firms take input prices as given. For simplicity, assume that prices are pre-determined before productivity $A_{i,t}$ realizes.

The firm produces its output $y_{i,t}$ using the following production function:

$$y_{i,t} = A_{i,t} n_{i,t}^{\alpha_{i,t}} \underbrace{\prod_{j=1}^N x_{ij,t}^{\omega_{ij,t}}}_{\equiv \tilde{y}_{i,t}} \quad (2.5)$$

with $\alpha_{i,t} + \sum_{j=1}^N \omega_{ij,t} = 1$, where $A_{i,t}$ is the stochastic productivity

$$\log A_{i,t} = \bar{a}_i + \sigma_{i,t-1} \epsilon_{i,t}. \quad (2.6)$$

We look at the components of productivity in (2.6) one by one. $\bar{a}_i$ denotes the sector-specific average productivity. $\sigma_{i,t}$ denotes the standard deviation scale of log productivity. $\epsilon_{i,t} \sim \mathcal{N}(0, 1)$ with joint normal distribution $\epsilon_t = (\epsilon_{1,t}, \epsilon_{2,t}, \dots, \epsilon_{N,t})' \sim \mathcal{N}(\mathbf{0}, \Sigma_{\epsilon,t})$ denotes the number of units of standard deviation.

Assume that uncertainty shocks are independent, i.e., $\sigma_{i,t}$'s are independent among sectors and are independent of level shocks $\epsilon_{i,t}$. Assume that the uncertainty shocks follow:

$$\log \sigma_{i,t}^2 = (1 - \phi) \log \bar{\sigma}_i^2 + \phi \log \sigma_{i,t-1}^2 + v_{i,t}, \quad (2.7)$$

where $\phi \in (0, 1)$ and $v_{i,t} \sim \mathcal{N}(0, \sigma_v^2)$ is the uncertainty shock. This decomposition isolates the uncertainty shock from the level shock: a shock to $\epsilon_{i,t}$ changes the realized productivity but not the distribution, while a shock to $\sigma_{i,t}$ changes the future conditional distribution of productivity. To be more precise, since $\sigma_{i,t} \perp \epsilon_{i,t}$,

$$\text{Var}_{t-1}(\log A_{i,t}) = \text{Var}_{t-1}(\sigma_{i,t-1} \epsilon_{i,t}) = \sigma_{i,t-1}^2.$$

The following lemma provides the closed-form solution to the conditional mean and standard deviation of productivity.

Lemma 2.1 (Conditional Mean and Standard Deviation of Productivity). *Given the information at the end of period $t-1$, productivity at period t is log-normally distributed with the conditional mean and standard deviation given by*

$$\mu_{i,t|t-1}^A = e^{\bar{a}_i + \frac{1}{2}\sigma_{i,t-1}^2}, \quad \sigma_{i,t|t-1}^A = \mu_{i,t|t-1}^A \sqrt{e^{\sigma_{i,t-1}^2} - 1}.$$

Proof. Conditional on the information at the end of period $t - 1$, (2.7) implies that the log of productivity has mean $\bar{a}_i$ and standard deviation $\sigma_{i,t-1}$. The remaining follows from the properties of log-normal distribution. $\square$

At the beginning of each period, each firm i faces labour costs $W_t n_{i,t}$ and intermediate input costs $\sum_{j=1}^N p_{j,t} x_{ij,t}$. Therefore, it needs to borrow $b_{i,t}$ from the financial intermediary to cover the costs:

$$W_t n_{i,t} + \sum_{j=1}^N p_{j,t} x_{ij,t} \leq b_{i,t}, \quad (2.8)$$

subject to a borrowing constraint, which will be discussed soon in the Section 2.1.3. At the end of the period, the firm has a revenue $R_{i,t} = p_{i,t} y_{i,t}$ and repays the debt $b_{i,t}$ together with interest at the intra-period loan rate $i_{i,t}^L$. Its nominal profit

$$\Pi_{i,t}^y = \max \left\{ R_{i,t} - \left(1 + i_{i,t}^L \right) b_{i,t}, 0 \right\}$$

will be distributed to the households.

2.1.3 Financial Intermediary

Competitive financial intermediaries take the deposit rate as tied to the policy rate $i_t^D = i_t$, so deposits D_t earn i_t in period $t + 1$. Period- t working-capital loans are funded by outstanding nominal deposits D_{t-1} and are priced off the predetermined funding rate i_{t-1} using a common risk-pricing rule:

$$i_{i,t}^L = i_{t-1} + s_{i,t},$$

where $s_{i,t}$ is the credit spread satisfying

$$s_{i,t} = \bar{s} + \psi \left(\sigma_{i,t|t-1}^R \right)^2, \quad (2.9)$$

for some $\psi > 0$, where

$$\sigma_{i,t|t-1}^R = \sqrt{\text{Var}_{t-1} (R_{i,t})} = p_{i,t} \tilde{y}_{i,t} \sigma_{i,t|t-1}^A$$

is the conditional standard deviation of firm revenue.

The intermediary's nominal balance-sheet constraint is

$$\sum_{i=1}^N b_{i,t} \leq D_{t-1}. \quad (2.10)$$

Its nominal profit Π_t^f is given by

$$\Pi_t^f = \sum_{i=1}^N \min \{R_{i,t}, (1 + i_{i,t}^L) b_{i,t}\} - (1 + i_{t-1}) D_{t-1}.$$

For each firm i , the intermediary imposes the borrowing constraint

$$b_{i,t} \leq \kappa \mu_{i,t|t-1}^R \exp \left(-\theta \sigma_{i,t-1}^2 \right), \quad (2.11)$$

where

$$\mu_{i,t|t-1}^R = \mathbb{E}_{t-1} [R_{i,t}] = p_{i,t} \tilde{y}_{i,t} \mu_{i,t|t-1}^A,$$

is the mean of revenue conditional on information available at the end of period $t - 1$. The parameter $\kappa > 0$ captures the pledgeability of expected revenue, and $\theta > 0$ captures the sensitivity of credit capacity to revenue risk. This constraint prevents the intermediary from severe credit risk, but it cannot eliminate default.

We will not explicitly solve for the financial intermediary's optimization problem in the benchmark model. We show that the credit-spread rule and the borrowing constraint can be interpreted as a reduced-form representation of competitive risk-based lending.

The financial intermediary's optimization problem is to maximize its nominal profit subject to a default-tolerance rule:

$$\max_{b_{i,t}} \mathbb{E}_{t-1} \Pi_t^f,$$

subject to

$$\mathbb{P}_{t-1} \left(R_{i,t} < (1 + i_{i,t}^L) b_{i,t} \right) \leq \delta,$$

for some $\delta \in (0, 1)$.

We first derive the implication of the default-tolerance rule. Note that $R_{i,t} = p_{i,t}\tilde{y}_{i,t}A_{i,t}$. Therefore, the rule implies that

$$\begin{aligned}\mathbb{P}_{t-1}\left(R_{i,t} < (1 + i_{i,t}^L)b_{i,t}\right) &= \mathbb{P}_{t-1}\left(\log A_{i,t} < \log\left((1 + i_{i,t}^L)b_{i,t}\right) - \log p_{i,t}\tilde{y}_{i,t}\right) \\ &= \Phi\left(\frac{\log\left((1 + i_{i,t}^L)b_{i,t}\right) - \log p_{i,t}\tilde{y}_{i,t} - \bar{a}_i}{\sigma_{i,t-1}}\right) \\ &\leq \delta.\end{aligned}$$

This is equivalent to

$$b_{i,t} \leq \frac{\mu_{i,t|t-1}^R}{1 + i_{i,t}^L} \exp\left(-\frac{1}{2}\sigma_{i,t-1}^2 - z_{1-\delta}\sigma_{i,t-1}\right) = \frac{\exp\left(-\frac{1}{2}\sigma_{i,t-1}^2\right)}{1 + i_{i,t}^L} \mu_{i,t|t-1}^R \exp(-z_{1-\delta}\sigma_{i,t-1}).$$

Denote $\kappa := \frac{\exp\left(-\frac{1}{2}\sigma_{i,t-1}^2\right)}{1 + i_{i,t}^L}$ and $\theta := z_{1-\delta}$. Then the default-tolerance rule implies the borrowing constraint (2.11).

Now consider the objective function:

$$\mathbb{E}_{t-1}\Pi_t^f = \sum_{i=1}^N \left[p_{i,t}\mu_{i,t|t-1}^A \tilde{y}_{i,t} (1 - \Phi(q_2)) + (1 + i_{i,t}^L)b_{i,t}\Phi(q_1) \right] - (1 - i_{t-1})D_{t-1},$$

where

$$\begin{aligned}q_1 &= \frac{\bar{a}_i - \log\left((1 + i_{i,t}^L)b_{i,t}\right) + \log(p_{i,t}\tilde{y}_{i,t})}{\sigma_{i,t-1}}, \\ q_2 &= \frac{\bar{a}_i + \sigma_{i,t-1}^2 - \log\left((1 + i_{i,t}^L)b_{i,t}\right) + \log(p_{i,t}\tilde{y}_{i,t})}{\sigma_{i,t-1}} = q_1 + \sigma_{i,t-1}.\end{aligned}$$

Since financial intermediaries are perfectly competitive, they should have zero expected profit for each loan i made. A footnote here is that the potential realized loss will be borne by household consumption. Suppose that the default-tolerance constraint binds. Then we have

$$(1 + i_{i,t}^L)b_{i,t} = p_{i,t}\mu_{i,t|t-1}^A \tilde{y}_{i,t} \exp\left(-\frac{1}{2}\sigma_{i,t-1}^2 - z_{1-\delta}\sigma_{i,t-1}\right).$$

This condition and the zero-expected profit condition for loan i together imply that

$$p_{i,t}\mu_{i,t|t-1}^A \tilde{y}_{i,t} (1 - \Phi(z_{1-\delta} + \sigma_{i,t-1})) + (1 + i_{i,t}^L)b_{i,t}(1 - \delta) = (1 + i_{t-1})b_{i,t}.$$

Further simplifying the expression yields

$$1 + i_{i,t}^L = (1 + i_{t-1}) \left[\exp \left(\frac{1}{2} \sigma_{i,t-1}^2 + z_{1-\delta} \sigma_{i,t-1} \right) (1 - \Phi(z_{1-\delta} + \sigma_{i,t-1})) + (1 - \delta) \right]^{-1}$$

We now show that the multiplier to $1 + i_{t-1}$ is greater than or equal to 1. Note that

$$\begin{aligned} & \exp \left(\frac{1}{2} \sigma_{i,t-1}^2 + z_{1-\delta} \sigma_{i,t-1} \right) (1 - \Phi(z_{1-\delta} + \sigma_{i,t-1})) \\ &= \exp \left(\frac{1}{2} \sigma_{i,t-1}^2 + z_{1-\delta} \sigma_{i,t-1} \right) \int_{z_{1-\delta} + \sigma_{i,t-1}}^{+\infty} \phi(u) du \\ &= \exp \left(\frac{1}{2} \sigma_{i,t-1}^2 + z_{1-\delta} \sigma_{i,t-1} \right) \int_{z_{1-\delta}}^{+\infty} \phi(x + \sigma_{i,t-1}) dx \\ &= \int_{z_{1-\delta}}^{+\infty} \phi(x) \exp(-\sigma(x - z_{1-\delta})) dx \\ &\leq \int_{z_{1-\delta}}^{+\infty} \phi(x) dx \\ &= \delta, \end{aligned}$$

where the inequality is derived from $x \geq z_{1-\delta}$. The result follows.

Taking log approximation on both sides, we get

$$i_{i,t}^L = i_{t-1} + \left[\delta - \exp \left(\frac{1}{2} \sigma_{i,t-1}^2 + z_{1-\delta} \sigma_{i,t-1} \right) (1 - \Phi(z_{1-\delta} + \sigma_{i,t-1})) \right].$$

It remains to show that the second term on the right-hand side is increasing in $\sigma_{i,t-1}$. Differentiating the second term with respect to $\sigma_{i,t-1}$, we get

$$\begin{aligned} & \frac{d}{d\sigma_{i,t-1}} \left[\delta - \exp \left(\frac{1}{2} \sigma_{i,t-1}^2 + z_{1-\delta} \sigma_{i,t-1} \right) (1 - \Phi(z_{1-\delta} + \sigma_{i,t-1})) \right] \\ &= \exp \left(\frac{1}{2} \sigma_{i,t-1}^2 + z_{1-\delta} \sigma_{i,t-1} \right) \left[\phi(z_{1-\delta} + \sigma_{i,t-1}) - (z_{1-\delta} + \sigma_{i,t-1}) (\Phi(-z_{1-\delta} - \sigma_{i,t-1})) \right]. \end{aligned}$$

Consider the function $f(a) = \phi(a) - a\Phi(-a)$. Note that

$$\begin{aligned} f'(a) &= \phi'(a) - \Phi(-a) + a\phi(-a) \\ &= -a\phi(a) - \Phi(-a) + a\phi(a) \\ &= -\Phi(-a) < 0. \end{aligned}$$

Since $\inf f(a) = f(+\infty) = 0 - 0 \cdot +\infty \leq 0$, we have $f(a) > 0$ for all $a \in \mathbb{R}$. This implies that the second term on the right-hand side is increasing in $\sigma_{i,t-1}$. Then we are able to define

the second term as $s_{i,t} = \bar{s} + \psi \text{Var}_{t-1}(R_{i,t})$ in a reduced-form representation of competitive risk-based lending.

2.1.4 Monetary Authority

The monetary authority sets the nominal interest rate i_t at the beginning of each period. This rate applies from the beginning of period $t + 1$ to its end, equivalently, deposits D_t earn i_t in period $t + 1$. Therefore, period- t loans are priced off the predetermined funding rate i_{t-1} through

$$i_{i,t}^L = i_{t-1} + s_{i,t}. \quad (2.12)$$

In the benchmark model, the nominal interest rate process is exogenous. We introduce an endogenous policy rule when discussing policy implications.

2.2 Agents' Decisions

2.2.1 Households

Households choose consumption C_t , labour supply N_t , and deposits D_t to maximize their discounted lifetime utility:

$$\max_{C_t, N_t, D_t} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(C_t, N_t),$$

where $\beta \in (0, 1)$ is the discount factor, subject to the budget constraint (2.4). The first-order conditions (FOC) imply the Euler equation (2.13) and the labour supply (2.14):

$$1 = \beta \mathbb{E}_t \left[\frac{1 + i_t^D}{1 + \pi_{t+1}} \frac{C_t}{C_{t+1}} \right], \quad (2.13)$$

$$\frac{W_t}{P_t} = \chi C_t N_t^\varphi. \quad (2.14)$$

2.2.2 Firms

Firms choose labour $n_{i,t}$, intermediate inputs $\{x_{ij,t}\}_{j=1}^N$ and debt $b_{i,t}$ to maximize their nominal profit $\Pi_{i,t}^y$:

$$\max_{n_{i,t}, \{x_{ij,t}\}_{j=1}^N, b_{i,t}} \mathbb{E}_{t-1} \Pi_{i,t}^y,$$

subject to the production function (2.5), the financial constraint (2.8) with Lagrange Multiplier $\lambda_{i,t}^1$ and the borrowing constraint (2.11) with Lagrange Multiplier $\lambda_{i,t}^2$.

Expanding using the definition of $\Pi_{i,t}^y$ and $\tilde{y}_{i,t}$, the objective function can be rewritten as

$$\mathbb{E}_{t-1} \Pi_{i,t}^y = \mu_{i,t|t-1}^A p_{i,t} \tilde{y}_{i,t} \Phi(q_2) - (1 + i_{i,t}^L) b_{i,t} \Phi(q_1),$$

where $\Phi(\cdot)$ is the cumulative distribution function of the standard normal distribution with probability density function $\phi(\cdot)$ and q_1 and q_2 are defined as in the proof of Lemma 2.1. Note that

$$\frac{\phi(q_2)}{\phi(q_1)} = \exp\left(-\sigma_{i,t-1} q_1 - \frac{\sigma_{i,t-1}^2}{2}\right) = \frac{(1 + i_{i,t}^L) b_{i,t}}{\mu_{i,t|t-1}^A p_{i,t} \tilde{y}_{i,t}}.$$

We can simplify the FOCs as follows

The FOCs are given by

$$\begin{aligned} [n_{i,t}] : \mu_{i,t|t-1}^A \alpha_{i,t} \frac{\tilde{y}_{i,t}}{n_{i,t}} \Phi(q_2) &= \lambda_{i,t}^1 \frac{W_t}{p_{i,t}} - \lambda_{i,t}^2 \kappa \mu_{i,t|t-1}^A \exp\left(-\theta \sigma_{i,t-1}^2\right) \alpha_{i,t} \frac{\tilde{y}_{i,t}}{n_{i,t}}; \\ [x_{ij,t}] : \mu_{i,t|t-1}^A \omega_{ij,t} \frac{\tilde{y}_{i,t}}{x_{ij,t}} \Phi(q_2) &= \lambda_{i,t}^1 \frac{p_{j,t}}{p_{i,t}} - \lambda_{i,t}^2 \kappa \mu_{i,t|t-1}^A \exp\left(-\theta \sigma_{i,t-1}^2\right) \omega_{ij,t} \frac{\tilde{y}_{i,t}}{x_{ij,t}}, \quad \forall j; \\ [b_{i,t}] : (1 + i_{i,t}^L) \Phi(q_1) &= \lambda_{i,t}^1 - \lambda_{i,t}^2. \end{aligned} \quad (2.15)$$

The first two FOCs implies that the cost-share conditions hold:

$$\frac{W_t n_{i,t}}{p_{j,t} x_{ij,t}} = \frac{\alpha_{i,t}}{\omega_{ij,t}}, \quad \forall j, \quad (2.16)$$

$$\frac{p_{j,t} x_{ij,t}}{p_{k,t} x_{ik,t}} = \frac{\omega_{ij,t}}{\omega_{ik,t}}, \quad \forall j \neq k. \quad (2.17)$$

The condition (2.15) implies the following Lemma.

Lemma 2.2 (Binding Working-Capital Constraint). *The working-capital constraint (2.8) binds in any interior solution with finite q_1 .*

Proof. From the FOC with respect to $b_{i,t}$,

$$\lambda_{i,t}^1 - \lambda_{i,t}^2 = (1 + i_{i,t}^L)\Phi(q_1).$$

Since $1 + i_{i,t}^L > 0$ and $\Phi(q_1) > 0$ for finite q_1 , we have

$$\lambda_{i,t}^1 = \lambda_{i,t}^2 + (1 + i_{i,t}^L)\Phi(q_1) > 0.$$

By complementary slackness for the working-capital constraint,

$$\lambda_{i,t}^1 \left[b_{i,t} - W_t n_{i,t} - \sum_{j=1}^N p_{j,t} x_{ij,t} \right] = 0.$$

Since $\lambda_{i,t}^1 > 0$, it follows that

$$b_{i,t} = W_t n_{i,t} + \sum_{j=1}^N p_{j,t} x_{ij,t}.$$

□

2.3 Equilibrium

There are three market in the economy: the goods market, the labour market, and the financial market. The goods market clears when

$$y_{i,t} = c_{i,t} + \sum_{j=1}^N x_{ji,t}, \quad \forall i. \quad (2.18)$$

This means that the total output of firm i is equal to the final consumption of the household plus the intermediate inputs used by other firms.

The labour market clears when

$$N_t = \sum_{i=1}^N n_{i,t}. \quad (2.19)$$

The financial market clears when

$$\sum_{i=1}^N b_{i,t} = D_{t-1}. \quad (2.20)$$

Given initial nominal deposits D_{-1} , initial money supply M_{-1}^S , and exogenous policy processes $\{i_t, X_t\}_{t=0}^\infty$, a competitive equilibrium is a stochastic sequence of allocations

$$\left\{ C_t, \{c_{i,t}\}_{i=1}^N, N_t, D_t, \{n_{i,t}\}_{i=1}^N, \{x_{ij,t}\}_{i,j=1}^N, \{y_{i,t}\}_{i=1}^N, \{b_{i,t}\}_{i=1}^N \right\}_{t=0}^\infty,$$

prices and rates

$$\left\{ P_t, \{p_{i,t}\}_{i=1}^N, W_t, i_t^D, \{i_{i,t}^L\}_{i=1}^N, \{s_{i,t}\}_{i=1}^N \right\}_{t=0}^\infty,$$

and profits and policy variables

$$\left\{ \Pi_t^y, \Pi_t^f, M_t^S, X_t \right\}_{t=0}^\infty$$

such that, for all t and all states:

- (a) Households maximize expected lifetime utility subject to the nominal budget constraint.
- (b) Firms maximize expected nominal profits subject to the production function, the working-capital constraint, and the borrowing constraint.
- (c) Financial intermediaries satisfy the balance-sheet constraint, the loan-pricing rule, and the borrowing-capacity rule.
- (d) The monetary authority sets i_t and X_t according to the benchmark policy process.
- (e) Goods, labor, and financial markets clear.

References

Acemoglu, Daron and Pablo D. Azar, “Endogenous Production Networks,” *Econometrica*, 2020, 88 (1), 33–82.